the doctrine of mutually assured destruction. But the danger of escalation in the age of AI is bigger, because cyber warfare is inherently different from nuclear warfare.

First, cyber weapons are much more versatile than nuclear bombs. Cyber weapons can bring down a country's electric grid, but they can also be used to destroy a secret research facility, jam an enemy sensor, inflame a political scandal, manipulate elections or hack a single smartphone. And they can do all that stealthily. They don't announce their presence with a mushroom cloud and a storm of fire, nor do they leave a visible trail from launchpad to target. Consequently, at times it is hard to know if an attack even occurred or who launched it. If a database is hacked or sensitive equipment is destroyed, it's hard to be sure who to blame. The temptation to start a limited cyberwar is therefore big, and so is the temptation to escalate it. Rival countries like Israel and Iran or the United States and Russia have been trading cyber blows for years, in an undeclared but escalating war.[38] This is becoming the new global norm, amplifying international tensions and pushing countries to cross one red line after another.

A second crucial difference concerns predictability. The Cold War was like a hyper-rational chess game, and the certainty of destruction in the event of nuclear conflict was so great that the desire to start a war was correspondingly small. Cyber warfare lacks this certainty. Nobody knows for sure where each side has planted its logic bombs, Trojan horses and malware. Nobody can be certain whether their own weapons would actually work when called upon. Would Chinese missiles fire when the order is given, or perhaps the Americans have hacked them or the chain of command? Would American aircraft carriers function as expected, or would they perhaps shut down mysteriously or sail around in circles?[39]

Such uncertainty undermines the doctrine of mutually assured destruction. One side might convince itself – rightly or wrongly – that it can launch a successful first strike and avoid massive retaliation. Even worse, if one side thinks it has such an opportunity, the temptation to launch a first strike could become irresistible, because one never knows how long the window of opportunity will remain open. Game theory posits that the most dangerous situation in an arms race is when one side feels it has an advantage but that this advantage is slipping away.[40]

Even if humanity avoids the worst-case scenario of global war, the rise of new digital empires could still endanger the freedom and prosperity of